

Designing for the Voiceless: Bridging Perception and Compassion in Pet Fish Care

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Abstract

Pet fish are among the most common companion animals worldwide, yet their well-being often goes unnoticed due to communication barriers and limited visibility of health conditions. Through field observations and user interviews, we found that high mortality rates in pet fish are frequently overlooked or misunderstood by owners. To address this issue, we present a novel system that leverages computer vision and augmented reality (AR) to visualize the health status of pet fish in real time. Our approach not only makes invisible health indicators perceptible but also uses AR-based visualizations on a mobile app to foster emotional connections and empathetic care between humans and aquatic pets. By transforming abstract data into engaging, interactive experiences, our design encourages more attentive and informed caregiving practices. This work contributes to the growing interest in designing for non-human agents and offers insights into how technology can support empathetic interactions across species.

CCS Concepts

• **Human-centered computing** → **Mixed / augmented reality**; **Collaborative interaction**; *Visual analytics*; *Visualization toolkits*.

Keywords

Computer Vision; Human-animal Interaction; Augmented Reality

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1 Introduction

Pet fish are among the most widely kept companion animals globally, with over 140 million households in the United States alone owning at least one aquarium or fish tank [15]. Unlike more socially expressive pets such as dogs or cats, however, fish do not exhibit clear behavioral cues that are easily interpretable by humans, often leaving their health conditions unnoticed until it is too late.

*Both authors contributed equally to this research.

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This "voicelessness" creates a significant gap between caretakers and their aquatic companions, especially among novice owners who may lack the knowledge or experience to detect early signs of illness or environmental stressors. Studies have shown that improper care due to misinterpretation of fish behavior is a leading cause of preventable mortality in home aquariums [5, 17]. Moreover, fish have their own language and signals, but these are often undetectable or misunderstood by humans, leading to their welfare being overlooked in discussions of responsible pet ownership and human-animal relationships.

To bridge this gap, we explore how visual representation methods can make invisible health indicators of pet fish visible and meaningful to their caretakers. Building on insights from visualization techniques and augmented reality (AR), we propose an interactive system that bases on computer vision to monitor fish behavior and AR to present real-time health information in an engaging and intuitive way. Our goal is to support more informed and empathetic caregiving practices for a group of animals that are both widely kept and frequently misunderstood.

Our work investigates the following three research questions:

- (1) Visualization: How can visualized health cues help novice fish owners recognize early signs of stress or illness in their pets?
- (2) Emotional Connection (Empathy Building): How can affective visualization and gamified feedback enhance emotional engagement with non-expressive animals like fish?
- (3) Targeted Guidance: How might interactive systems provide targeted guidance to encourage more consistent and responsible care behaviors among pet fish owners?

By combining computer vision, AR, and game design, our project reimagines how humans can engage with and care for non-expressive companion animals, offering new possibilities for inclusive, affective, and tech-mediated interspecies relationships. Our work makes two creative contributions:

- (1) Behavior-based Health Inference: Unlike previous systems that primarily rely on appearance or physiological indicators, our work establishes a technical framework for health status recognition based on the swimming trajectory patterns of pet fish, enabling the early detection of illness through changes in movement behavior.
- (2) Empathetic AR Game Design: We introduce a mobile AR app that transforms abstract health data into an emotionally resonant experience through gamified visualizations, helping users build stronger emotional connections with their fish.

2 Related Works

2.1 Technologies for Detecting Fish Health Status

Recent advances in computer vision and machine learning have enabled automated monitoring of fish behavior and health, particularly in aquaculture and laboratory settings [2, 3, 6, 10, 11, 14]. Several studies have focused on detecting disease, or abnormal behaviors using video-based tracking and motion analysis [4, 12, 13, 16, 19]. For example, Liang et al. [11] used computer vision to identify the health status of guppies, and enhanced the model's ability to recognize subtle differences to achieve overall health monitoring of the fish population.

Similarly, Zhou et al. applied real-time object detection based on YOLOv3 for the diagnosis of *Epinephelus coioides* nodavirus disease in large yellow croaker (*Larimichthys crocea*) [20]. These systems often rely on features like swimming speed, directionality, and inter-fish distances to infer health conditions. However, most are designed for large-scale environments with controlled lighting and camera setups, making them unsuitable for home aquariums. Existing methods focusing on appearance-based indicators—such as color changes or physical lesions—are also affected by environmental factors and may not detect early-stage health issues. Given that a significant proportion of pet fish are kept in home aquariums, there is a need to develop systems that integrate detection methods with user-accessible devices like smartphones or standard webcams.

2.2 Augmented Reality for Biological Data Visualization and Complex Skill Acquisition

A significant gap exists between data identification, visualization, and the effective communication of insights in a user-comprehensible manner. While accurate detection of fish health is crucial, translating this information into meaningful insights for everyday users remains a challenge [8, 18].

Augmented reality (AR) has shown great potential in bridging this gap by transforming abstract or technical data into intuitive visual representations. Weng et al. [18] demonstrated that AR technology has the potential to improve students' analytical-level learning outcomes and foster more positive attitudes toward biological education. Similarly, other studies have explored the use of AR in life science education and medical training, where it has been used to visualize complex biological structures and health status of pet in an interactive manner [1]. In the study by Li et al., "Design and Evaluation of an Augmented Reality-Based Pet Care Instructional System", teacher feedback emphasized that the AR system significantly enhanced student engagement and improved learning efficiency [9]. These applications highlight AR's ability to support not only data visualization but also the development of complex cognitive and motor skills. By integrating AR into fish health monitoring systems, we aim to enable non-expert users to understand pets' health needs.

3 Design Process

3.1 User Research and Pain Point Identification

To inform the design of our AR-based fish health visualization system, we conducted preliminary user research from March 2025 to May 2025. This included in-depth interviews with three experienced fish hobbyists and an online survey completed by 24 novice and experienced fish keepers. The participants of online survey were nearly balanced in gender, consisting of 11 male and 13 female respondents. In terms of age distribution, two main groups were identified: young adults aged 20–30 and middle-aged to senior individuals aged 50–60. Regarding fish-keeping experience, approximately half of the participants (10 out of 24) were beginners with less than one year of experience, while the other half (14 participants) had more than two years of experience, with some exceeding five years. This diverse sample ensured a comprehensive understanding of both novice and experienced users' needs and practices, which informed the usability and functional design of the system. The interviewees included a 23-year-old male novice, a 50-year-old male veteran with four years of experience and a 23-year-old female with less than a year of experience. The interview outline covered the following topics: 1) current practices and challenges in monitoring fish health and well-being, 2) emotional relationship and bonding between human and pet fish, 3) expectations for technology-supported tools in fish care.

From our conversations with experienced fish keepers, we learned that while they emphasize the importance of providing essential physical resources—such as tanks, filters, and food—they also attempt to support each other through shared knowledge, including images and descriptions of fish behaviors. However, many noted a recurring challenge: new owners often struggle to interpret fish behavior or recognize early signs of distress. When asked why more people are turning to fish over traditional pets like dogs and cats, one owner shared: "They fit my busy lifestyle. Watching them swim helps me unwind, especially after a long day at work." Interviews with hobbyists revealed similar themes. One participant explained how they monitor their fish's emotional and health status: "I'm still learning, but if something seems wrong, I consult local fish merchants for advice. And I keep an eye on their behavior and make sure they're eating well." This highlights the reliance on informal knowledge networks and the lack of accessible tools for monitoring.

Our broader online research further identified two dominant demographics among fish keepers: (1) middle-aged and older adults who appreciate the calming presence of fish, and (2) young professionals and students who seek low-maintenance companionship due to time constraints. Synthesizing these insights, we developed user personas to represent key stakeholder groups and distilled the following pain points:

- **Lack of Physical Feedback and Emotional Connection:** Fish cannot express discomfort or emotions in ways humans easily understand, leading to feelings of detachment and reduced caregiving motivation.
- **Cognitive Load and Declining Motivation:** Proper fish care involves nuanced knowledge of water chemistry, feeding habits, and behavioral cues, which can overwhelm beginners and lead to disengagement.

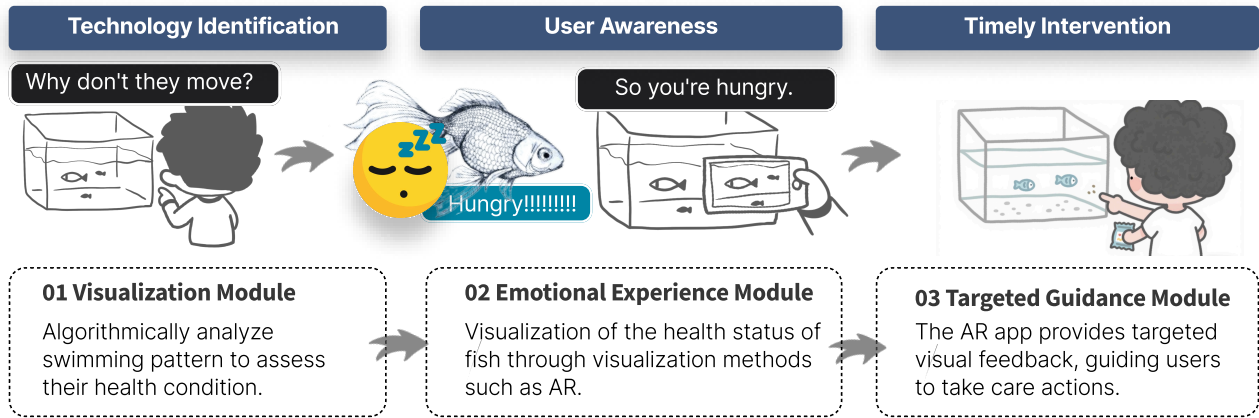


Figure 1: System functional architecture and technical architecture.

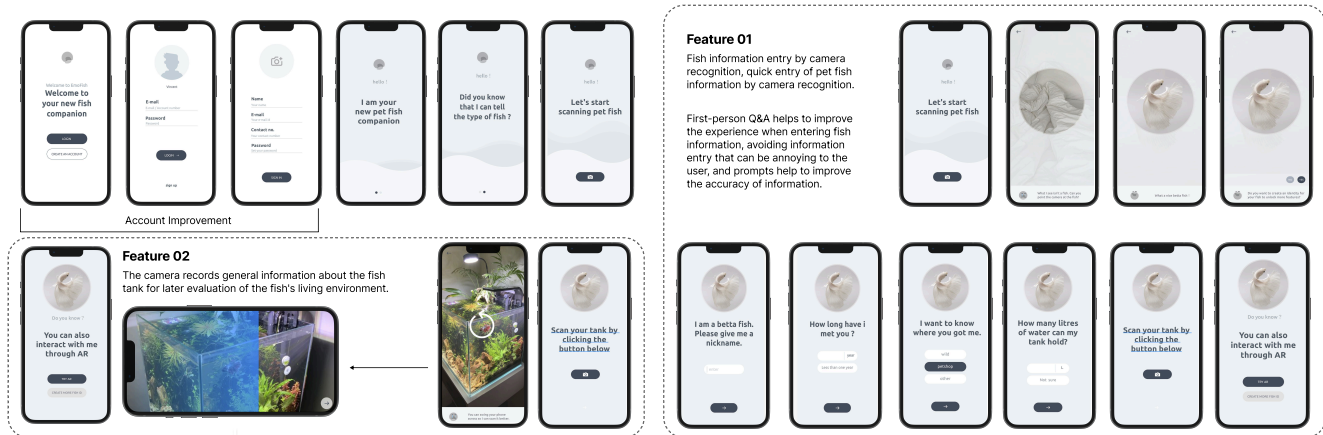


Figure 2: Pet fish and aquarium environment information input function interface framework.

- **Delayed Feedback Leading to Mortality:** Many fish deaths are preventable but occur due to misinterpretation of symptoms or delayed intervention.

These findings underscore the need for a supportive, intuitive, and emotionally engaging tool that bridges the gap between fish and their caretakers—an insight that directly informs our system's design direction. Based on users' repeated expressions of "inability to understand the state of the fish" and "lagging care feedback", we propose that the system must possess: spherical timing, low cognitive burden and emotional engagement, and large characteristics. Correspondingly, we divide the system functions into three parts: visual feedback, goal guidance, and emotional interaction, each addressing users' pain points respectively.

3.2 System Design

Building upon the insights from our user research, we formulated a central design question: How to make the process of keeping fish simple and fun? This question guides us to define the functional

positioning of the system, "simply" solving the technical identification of fish health conditions, and "interestingly" coming from the expectations of users in the interview for relaxed companionship and stress reduction. We focus on how users perceive and participate in this information to increase their emotional investment and enhance their long-term engagement. To achieve these two goals, we have identified three core functional modules that are consistent with user requirements and technical capabilities (Figure 1):

- (1) **Visualization Module:** This module uses computer vision (mainly OpenCV based on YOLOV8) to analyze the swimming patterns of fish, and sets indicators such as swimming frequency trajectory and speed as abnormal health features according to the ecological behavior standards in reference [7]. Insights are translated into intuitive visual cues—making health indicators accessible to users without prior knowledge.
- (2) **Targeted Guidance Module:** An AR-based system provides real-time, context-aware guidance by overlaying recommendations—like water quality alerts or feeding reminders—onto the live

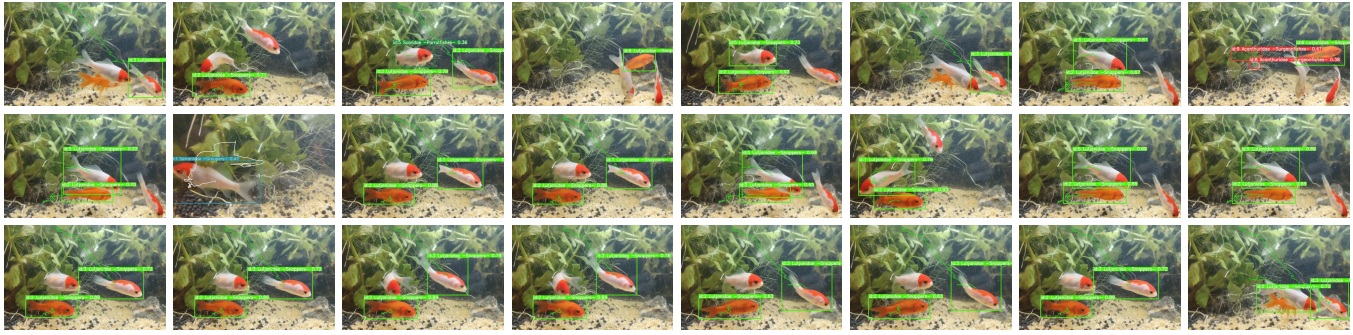


Figure 3: Pet fish identification and path tracking technology test process.

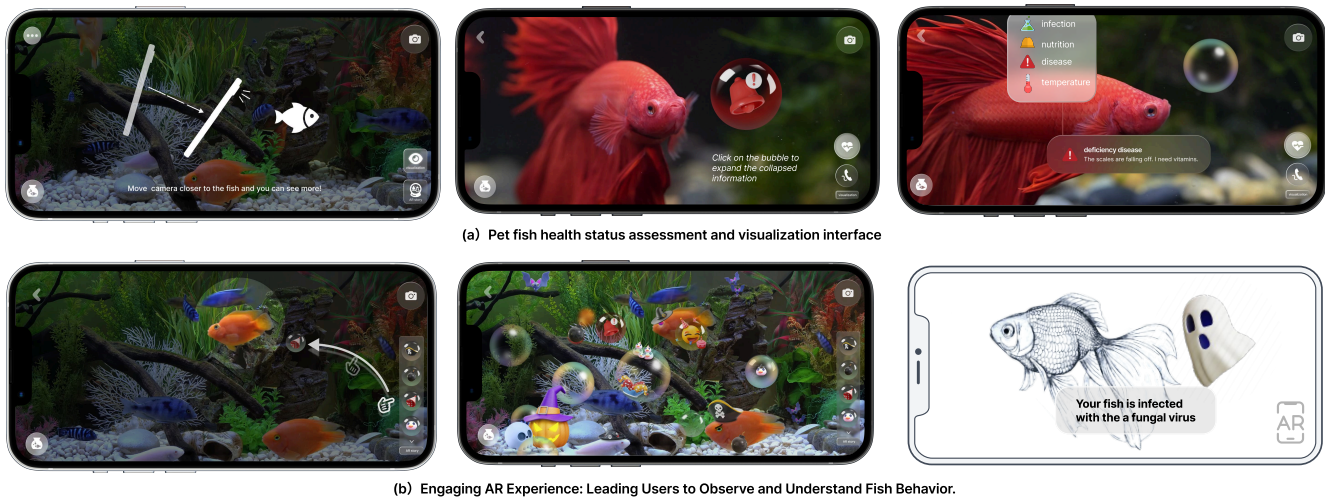


Figure 4: Interactive AR features for (a) health status display and (b) user engagement.

camera feed. Provide targeted guidance and suggestions based on specific situations. This helps users make informed decisions and take timely actions to improve tank conditions.

- (3) Emotional Experience Module: To build empathy and engagement, we use AR to anthropomorphize fish through playful elements like animated emojis. For example, a “sad” emoji appears when stress is detected, encouraging users to check on their fish and reinforcing emotional connection. These affective cues are embedded throughout the interface to enhance emotional connection and encourage consistent caregiving behaviors.

3.3 Game Design

The system integrates three core functional modules into five distinct interface components, designed to guide users through a seamless and engaging fish-keeping experience—from initial setup to daily interaction.

Fish Information Entry via Camera Recognition (Feature 01): This feature enables users to quickly input basic information about their pet fish using camera-based image recognition. Upon first launch,

users can take a photo of their fish, and the app automatically identifies species, size, and other relevant attributes. To enhance accuracy and reduce the burden of manual data entry, a first-person Q&A format guides users through additional inputs—such as tank type or expected lifespan—making the onboarding process intuitive and user-friendly (Figure 2).

Tank Environment Capture via Camera Input (Feature 02): In addition to fish identification, the app uses the device’s camera to capture visual data about the aquarium environment. Users are prompted to photograph the tank from multiple angles, allowing the system to estimate its size, layout, and decorative elements. This contextual data supports more accurate environmental evaluations and helps generate personalized care suggestions tailored to the specific living conditions of the fish (Figure 2).

Health and Emotional Status Assessment via Movement Analysis (Feature 03): Using computer vision techniques (OpenCV based on YOLOv8), the system analyzes real-time video footage of the fish to extract swimming patterns (Figure 3). After obtaining the complete trajectory, key behavioral features—such as swimming speed, acceleration, curvature, and static duration—were extracted.

Health status was inferred using heuristic rule models based on [7], where, for example, reduced swimming frequency, sudden changes in curvature, or drops in speed may indicate stress or sub-health. To enhance robustness against water impurities and other visual disturbances, the system incorporates background subtraction and adaptive thresholding. When potential health issues are detected, an AR warning bubble appears above the fish in the live camera view. Users can tap the bubble to access detailed explanations of the issue and receive tailored care recommendations, such as checking for signs of disease (Figure 4a).

AR-Based Emotional Interaction for User Engagement (Feature 04): This feature introduces an interactive AR game designed to foster emotional engagement between users and their fish (Figure 4b). Within the game, users can drag and drop special "effect spheres" from the right side of the interface onto specific fish in the tank view. Once applied, these spheres generate AR effects that follow the selected fish—such as a pirate hat or animated bubbles. When two fish with compatible effect spheres come close to each other in the AR view, the system triggers playful storyline dialogues, enhancing the sense of narrative and connection. Importantly, this is not merely a decorative feature. The availability of certain limited-effect spheres is tied to the user's caregiving performance. Additionally, some effect spheres serve as subtle visual indicators of fish health. For instance, a flickering sphere may signal stress, while a color change might reflect poor water quality. Through this gamified interaction, users receive real-time feedback on their fish's well-being and are gently prompted to take appropriate care actions—all within an engaging and emotionally resonant experience.

Proactive Care Reminders via Friendly Notifications (Feature 05): The system sends timely, friendly messages from the fish to remind users of important tasks such as feeding, water changes, and tank cleaning. These notifications are personalized based on the fish's needs and environmental conditions, and they simulate a conversational tone—creating a sense of companionship and responsibility. Each completed task may also unlock achievements or virtual rewards, reinforcing positive habits and enhancing long-term engagement.

4 Conclusion

This study presents the development and evaluation of an innovative AR-based fish care system that meets the needs of both novice and experienced aquarists. The system offers a promising approach to smart pet care, combining computer vision with gamified interaction. Future work will focus on improving recognition algorithms, customizing game designs for different age groups, and conducting extensive user testing to assess long-term usability. We will also introduce transparent data usage instructions, evaluating users' acceptance and emotional responses.

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